

pre-tRNA and mature-tRNA protocol

1.对数据进行初步质控，两种方式

```
##bbduk软件
bbduk.sh in=Prectr0923CTC-LGJ24582_L1_1.fq.gz in2=Prectr0923CTC-
LGJ24582_L1_2.fq.gz ref=adapter.fa out=CTC_trim_1.fq.gz out2=CTC_trim_2.fq.gz
trimq=30 minlength=50

##trim软件
trim_galore -q 30 --length 50 --illumina -paired ${n}_1.fq.gz ${n}_2.fq.gz
```

2.对质控完的数据，进行mature-tRNA比对

```
#!/bin/sh

# index="/home/jj2024/00.db/01.hg38/05.hg38_trna_illumina/01.mature_tRNA/hg38-
mature-tRNAs-CCA-cluster"
index="/home/jj2024/00.db/01.hg38/05.hg38_trna_illumina/01.mature_tRNA/03.addrRNA_
_snRNA_ivt3_dm_miala_ugc/hg38-mature-tRNAs-CCA-cluster"

for i in *.1_val_1.fq.gz;do
    j=${i/.1_val_1.fq.gz/}
    bowtie2 -p 8 --end-to-end --very-sensitive --dovetail --no-mixed --no-
discordant --fr -k 20 -x 1000 -x ${index} -1 ${j}.1_val_1.fq.gz -2
${j}.2_val_2.fq.gz | \
    samtools view -h -bF 4 - | \
    samtools sort -@ 4 - > ${j}.bam
done
```

3.对mature-tRNA进行reads计数

```
salmon quant -t
/home/jj2024/00.db/01.hg38/05.hg38_trna_illumina/01.mature_tRNA/03.addrRNA_snRNA_
ivt3_dm_miala_ugc/hg38-mature-tRNAs-CCA-cluster.fa -l A -a Prectr0923PIC-
LGJ24583_L1.bam --output 01.pic
```

4.先取未必对上mature-tRNA的数据，再进行下一步的pre-tRNA比对

```

#!/bin/sh

index="/home/jj2024/00.db/01.hg38/05.hg38_trna_illumina/01.mature_tRNA/03.addrRNA
_snRNA_ivt3_dm_miala_ugc/hg38-mature-tRNAs-CCA-cluster"

for i in *_1_val_1.fq.gz;do
    j=${i/_1_val_1.fq.gz/}
    bowtie2 -p 8 --end-to-end --very-sensitive --dovetail --no-mixed --no-
discordant --fr -k 20 -x 1000 -x ${index} -1 ${j}_1_val_1.fq.gz -2
${j}_2_val_2.fq.gz | \
    samtools view -h -bf 4 - | \
    samtools sort -@ 4 - > ${j}.bam
done

#samtools fastq -1 Prectr0923CTC-LGJ24582_L1_1.fq -2 Prectr0923CTC-
LGJ24582_L1_2.fq -n Prectr0923CTC-LGJ24582_L1.bam

```

5.pre-tRNA比对

```

salmon quant -i /home/jj2024/00.db/01.hg38/05.hg38_trna_illumina/02.Pre-
tRNA/08.add_50bp_rRNA_snRNA_ivt3_dm_miala_ugc/hg38-tRNAs-50bp -l A -1 <fq> -2
<fq> --output <name>

```